

RAED OUIRIEMMI

SOFTWARE DEVELOPER · DATA SCIENCE

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PROFESSIONAL SUMMARY

Data Science Master's student with a Bachelor's in Software Engineering and two years building production applications end to end — web and mobile front ends, Python back ends, and an AI component (LLMs, NLP, or computer vision). Comfortable owning a project from the first commit to the server it runs on, including Docker deployment and hardening on a self-managed VPS. Seeking a developer or data role in Canada.

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, C#, C++
Web & Mobile	React, React Native (Expo), Node.js, FastAPI, HTML, CSS, Tailwind
Machine Learning	TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Deep Learning / AI	PyTorch, Keras, YOLO, LLaMA 2, BART, Helsinki-NLP, OpenAI & OpenRouter APIs
Databases	PostgreSQL, MySQL, MSSQL, MongoDB, Firebase, Supabase (SQLAlchemy, Alembic)
DevOps & Tooling	Docker, Docker Compose, nginx, Linux/Ubuntu, UFW, fail2ban, Git, GitHub, Maven, JUnit
Methods	UML, Agile, automated testing, web scraping, i18n/localization (EN / FR / Arabic RTL)

PROFESSIONAL EXPERIENCE

AI / Full-Stack Engineer Intern — AKR Smart Consulting

Master's final-year project (PFE) · Feb 2026 – May 2026

- Built a multi-tenant **AI chatbot platform** (SaaS): admins create assistants trained on their own content — uploaded documents and crawled websites — via a **RAG pipeline** (sentence-transformers embeddings + ChromaDB vector search).
- Delivered a FastAPI + MongoDB back end, a React admin dashboard, an embeddable JavaScript chat widget, and a Celery / Redis / Playwright web crawler.
- JWT authentication with 2FA (TOTP), per-IP rate limiting, and Dockerized deployment.

Stack: FastAPI, MongoDB, React, ChromaDB, sentence-transformers, Celery, Redis, Playwright, Docker

Software Development Intern — AZ IT

Monastir, Tunisia · Feb 2024 – Aug 2024

- Delivered two projects back to back over a six-month internship: a document-sharing platform and a business-address web app.
- **Accountant-client mobile app (React Native):** built a mobile app and web dashboard letting accountants and clients exchange documents securely without email; added secure file upload and a single-view client dashboard.
- **Business-address platform (graduation project):** a web app letting small companies register a legal office address without renting physical space; built the React.js front end, Node.js back end, and the address-declaration workflow.

SELECTED PROJECTS

AppFullSport — Fitness Coaching App

2025 – 2026

- Full-stack mobile app: React Native client, FastAPI back end, and a React admin panel, running in Docker on a VPS I administer.
- Per-set weight/rep tracking, rest timer, automatic personal-record detection; seeded ~83 exercises. Cross-platform step counting (iOS HealthKit/CoreMotion, Android step sensor).
- Hardened the server: UFW + DOCKER-USER iptables, fail2ban, key-only SSH, account lockout, nginx + FastAPI rate limits, CSP/HSTS, rotated secrets, Alembic migrations.
- Fully localized EN / FR / Arabic (RTL) — including backend error messages via the Accept-Language header.

Stack: React Native (Expo SDK 54), FastAPI, SQLAlchemy, Alembic, PostgreSQL 16, Docker Compose, nginx, React, Tailwind

Natural-Language → SQL Database Chatbot

2025

- Web chatbot that turns plain-English questions into SQL against an MSSQL database and returns concise answers.

Stack: React, FastAPI, MSSQL, OpenRouter API

Discord Multi-Account Management Tool

2025

- Web app managing multiple Discord accounts via tokens and proxies, with a real-time admin dashboard; bots auto-reply using the OpenAI API and deliver messages through Discord webhooks.

Stack: React, FastAPI, Discord Webhooks, OpenAI API

IT Product-Search Chatbot (LLaMA 2)

2024

- LLaMA 2 chatbot that helps users find laptops, phones, and IT products, with budget- and use-case-based recommendations; a daily scraper refreshes the MyTek catalog.

Text Summarization & Translation Web App

2024

- FastAPI + React app using BART for summarization and Helsinki-NLP for translation across 10+ languages.

License-Plate Detection in Video (YOLO)

Sep 2024

- Trained a YOLO model to detect and extract vehicle license plates from video, frame by frame, in real time.

EDUCATION

M.Sc. Data Science

Faculty of Sciences, Monastir · Sep 2024 – Present

B.Sc. Software Engineering & Information Systems

Faculty of Sciences, Monastir · 2021 – 2024

Scientific Baccalaureate

Dhouiher High School, Zarzis · 2021

LANGUAGES

- **Arabic** — Native
- **French** — Upper-intermediate (TCF 400/699, B2)
- **English** — Intermediate (B1); work in English daily

STRENGTHS

Problem solving, critical thinking, adaptability, leadership, creativity, teamwork.